Lecture III – Accuracy

Overfitting

Overfitting results when a decision tree fits the data on which it was trained, but has decreased effectiveness when applied to other datasets. Overly complex decision trees are a sign of overtraining. Another approach to prevent overfitting is to partition the data into different cohorts. One cohort will be used to train the data. The other cohort will test the decision tree's ability to make correct predictions. We call one cohort training and the other testing.

Confusion Matrix

		Predicted	
Actual		**A**	**B**
	A	AA	AB
	B	BA	BB

The first value is what is actual, and the second is what is predicted. For example, the first cell, AA, indicates the number of times A was predicted correctly. The cell to the right, AB, is the count of how A's were predicted to be B's.

From these numbers, we can calculate:

Total = AA + AB + BA + BB

Accuracy = (AA+BB) / Total

True Positive Rate of A (aka Sensitivity or recall of A) = AA / (AA + AB)

True Positive Rate of B (aka Sensitivity or recall of B) = BB / (BB + BA)

False Positive Rate of A = BA / (BB + BA)

False Positive Rate of B = AB / (AA + AB)

Precision of A = AA / (AA+BA)

Precision of B = BB / (BB + AB)

Harmonic Mean (aka F Measure) = (2 X Precision X Recall) / (Precision + Recall)

When a target field is binary True/False, AA represents True Positive predictions, BB represents True Negative, AB represents False Negative, and BA represents False Positives.

Example

Predicted

Actual		**A**	**B**
	A	10	2
	B	20	8

Accuracy = (10 + 8) / 10 + 2 + 20 + 8 = 18 / 40 = .45

True Positive Rate of A (Sensitivity of A or recall) = 10 / 10 + 2 = 10 / 12 = .83333

False Positive Rate of A = 20 / 20 + 8 = .071

Precision of A = 10 / 10 + 20 = 10 / 30 = .33333

Harmonic Mean = (2 X .3333 X .83333) / (.3333 + .83333) = .5549 / 1.1663 = .47578

Ensembles

Ensembles are a training/testing technique that allows you to iteratively improve the performance of your model. It works with many different algorithms, like Decision Trees, Support Vector Machines, and Neural Nets.

Cross Validation

In this approach, the original dataset is separated into multiple buckets, or “folds”. The machine assigns each fold the role of training or testing and a model is built. Then the training/testing roles are reassigned and another model is built. This repeats until each fold has been used for testing and the model with the best results is selected.

Bagging

The original data set is randomly sampled multiple times to produce “bags”. Each bag is then used to produce a model. Each of these models votes of which class label should be assigned. The final class label is awarded based on which label received the most votes.

Boosting

Similar to bagging, multiple models are built based on different samples. However, boosting places greater emphasis on examples that are difficult to classify correctly. After a round of testing, the instances that were classified incorrectly are given greater representation in the next sample.

Bibliography

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